

Лабораторная работа №7

Управление журналами событий в системе

Лабси Мохаммед

2026-10-02

1. Цели и задачи работы
2. Мониторинг системных событий
3. Настройка rsyslog
4. Настройка отладочного журнала
5. Работа с journalctl
6. Постоянный журнал journald
7. Выводы по проделанной работе

1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Получить практические навыки работы с системными журналами Linux.

Изучить мониторинг событий в реальном времени, настройку **rsyslog**, работу с **journalctl**, фильтрацию системных сообщений и настройку постоянного хранения журнала **journald**.

2. 2. Мониторинг системных событий

2.1 Отслеживаю системный журнал

```
0x000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x58b2c)#012#8 0x00000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3b900)#012#9 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)#012#10 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)#012#012Stack trace of thread 3780:#012#0 0x00007f79c6cfe25d syscall (libc.so.6 + 0x10425d)#012#1 0x00000000043a812 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x3a812)#012#2 0x000000000458786 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x58786)#012#3 0x0000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x56d8)#012#4 0x00007f79c6c2458e __libc_start_call_main (libc.so.6 + 0x2a58e)#012#5 0x00007f79c6c24649 __libc_start_main@@GLIBC_2.34 (libc.so.6 + 0x2a649)#012#6 0x0000000004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x44fa)#012ELF object binary architecture: AMD x86-64
Oct 2 11:12:00 mohammedlabsi systemd[1]: systemd-coredump@19-3784-0.service: Deactivated successfully.
Oct 2 11:12:03 mohammedlabsi su[3771]: FAILED SU (to root) mohammedlabsi on pts/2
Oct 2 11:12:05 mohammedlabsi kernel: traps: VBoxClient[3795] trap int3 ip:41e7eb sp:7f79b85b4cd0 error:0 in VBoxClient[1e7eb, 400000+c8000]
Oct 2 11:12:05 mohammedlabsi systemd-coredump[3796]: Process 3792 (VBoxClient) of user 1000 terminated abnormally with signal 5/TRAP, processing...
Oct 2 11:12:05 mohammedlabsi systemd[1]: Started systemd-coredump@20-3796-0.service - Process Core Dump (PID 3796/UID 0).
Oct 2 11:12:06 mohammedlabsi systemd-coredump[3797]: Process 3792 (VBoxClient) of user 1000 dumped core.#012#012Module /opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient without build-id.#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.
```

Рисунок 1: Мониторинг системного журнала и фиксация неудачной попытки выполнения su

2.2 Проверяю работу команды logger

```
tions-7.2.16/bin/VBoxClient + 0x3a812)#012#2 0x0000000000458786 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x58786)
#012#3 0x00000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x56d8)#012#4 0x00007f79c6c2458e __libc_start
_call_main (libc.so.6 + 0x2a58e)#012#5 0x00007f79c6c24649 __libc_start_main@@GLIBC_2.34 (libc.so.6 + 0x2a649)#012#6 0x000000
00004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient + 0x44fa)#012ELF object binary architecture: AMD x86-64
Oct 2 11:12:41 mohammedlabsi systemd[1]: systemd-coredump@27-3878-0.service: Deactivated successfully.
Oct 2 11:12:42 mohammedlabsi mohammedlabsi[3884]: hello
Oct 2 11:12:43 mohammedlabsi mohammedlabsi[3889]: hello
Oct 2 11:12:46 mohammedlabsi kernel: traps: VBoxClient[3894] trap int3 ip:41e7eb sp:7f79b85b4cd0 error:0 in VBoxClient[1e7eb,
400000+c8000]
Oct 2 11:12:46 mohammedlabsi systemd-coredump[3895]: Process 3891 (VBoxClient) of user 1000 terminated abnormally with signal
5/TRAP, processing...
Oct 2 11:12:46 mohammedlabsi systemd[1]: Started systemd-coredump@28-3895-0.service - Process Core Dump (PID 3895/UID 0).
Oct 2 11:12:46 mohammedlabsi systemd-coredump[3895]: Process 3891 (VBoxClient) of user 1000 dumped core: #012#012Module /usr/lib/
```

Рисунок 2: Регистрация сообщения hello с помощью команды logger

2.3 Просматриваю журнал безопасности

```
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi# tail -n 20 /var/log/secure  
Oct  2 11:09:03 mohammedlabsi sshd[1285]: Server listening on :: port 22.  
Oct  2 11:09:03 mohammedlabsi (systemd)[1517]: pam_unix(systemd-user:session): session opened for user gdm(uid=42) by gdm(uid=0)  
Oct  2 11:09:03 mohammedlabsi gdm-launch-environment][1496]: pam_unix(gdm-launch-environment:session): session opened for user gdm(uid=42) by (uid=0)  
Oct  2 11:10:08 mohammedlabsi unix_chkpwd[2203]: password check failed for user (mohammedlabsi)  
Oct  2 11:10:08 mohammedlabsi gdm-password][2194]: pam_unix(gdm-password:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=mohammedlabsi  
Oct  2 11:10:08 mohammedlabsi gdm-password][2194]: gkr-pam: unable to locate daemon control file  
Oct  2 11:10:08 mohammedlabsi gdm-password][2194]: gkr-pam: stashed password to try later in open session  
Oct  2 11:10:13 mohammedlabsi gdm-password][2206]: gkr-pam: unable to locate daemon control file  
Oct  2 11:10:13 mohammedlabsi gdm-password][2206]: gkr-pam: stashed password to try later in open session  
Oct  2 11:10:13 mohammedlabsi (systemd)[2215]: pam_unix(systemd-user:session): session opened for user mohammedlabsi(uid=1000) by mohammedlabsi(uid=0)  
Oct  2 11:10:13 mohammedlabsi gdm-password][2206]: pam_unix(gdm-password:session): session opened for user mohammedlabsi(uid=1000) by mohammedlabsi(uid=0)  
Oct  2 11:10:13 mohammedlabsi gdm-password][2206]: gkr-pam: gnome-keyring-daemon started properly and unlocked keyring  
Oct  2 11:10:23 mohammedlabsi gdm-launch-environment][1496]: pam_unix(gdm-launch-environment:session): session closed for user gdm  
Oct  2 11:11:16 mohammedlabsi (systemd)[3487]: pam_unix(systemd-user:session): session opened for user root(uid=0) by root(uid=0)  
Oct  2 11:11:16 mohammedlabsi su[3474]: pam_unix(su:session): session opened for user root(uid=0) by mohammedlabsi(uid=1000)  
Oct  2 11:11:19 mohammedlabsi su[3535]: pam_unix(su:session): session opened for user root(uid=0) by mohammedlabsi(uid=1000)  
Oct  2 11:11:22 mohammedlabsi su[3577]: pam_unix(su:session): session opened for user root(uid=0) by mohammedlabsi(uid=1000)  
Oct  2 11:11:57 mohammedlabsi su[3577]: pam_unix(su:session): session closed for user root  
Oct  2 11:12:01 mohammedlabsi unix_chkpwd[3790]: password check failed for user (root)  
Oct  2 11:12:01 mohammedlabsi su[3771]: pam_unix(su:auth): authentication failure; logname=mohammedlabsi uid=1000 euid=0 tty=/dev/pts/2 ruser=mohammedlabsi rhost= user=root  
root@mohammedlabsi:/home/mohammedlabsi#
```

Рисунок 3: Просмотр последних записей журнала безопасности

3. 3. Настройка rsyslog

3.1 Устанавливаю и запускаю Apache

```
root@rocky:~# rpm -q httpd
httpd.x86_64 2.4.63-13.el10_2.6.x86_64

root@rocky:~# rpm -q --whatprovides httpd
apr-1.7.5-3.el10.x86_64
apr-util-1.6.3-23.el10_2.1.x86_64
apr-util-openssl-1.6.3-23.el10_2.1.x86_64
apr-util-ldap-1.6.3-23.el10_2.1.x86_64
httpd-2.4.63-13.el10_2.6.x86_64
httpd-core-2.4.63-13.el10_2.6.x86_64
httpd-filesystem-2.4.63-13.el10_2.6.noarch
httpd-tools-2.4.63-13.el10_2.6.x86_64
mod_http2-2.0.29-4.el10_2.2.x86_64
mod_lua-2.4.63-13.el10_2.6.x86_64
rocky-logos-httpd-100.5-3.el10.noarch

Installed:
apr-1.7.5-3.el10.x86_64          apr-util-1.6.3-23.el10_2.1.x86_64    apr-util-ldap-1.6.3-23.el10_2.1.x86_64
apr-util-openssl-1.6.3-23.el10_2.1.x86_64  httpd-2.4.63-13.el10_2.6.x86_64      httpd-core-2.4.63-13.el10_2.6.x86_64
httpd-filesystem-2.4.63-13.el10_2.6.noarch  httpd-tools-2.4.63-13.el10_2.6.x86_64  mod_http2-2.0.29-4.el10_2.2.x86_64
mod_lua-2.4.63-13.el10_2.6.x86_64          rocky-logos-httpd-100.5-3.el10.noarch

Complete!
root@mohammedlabsi:/home/mohammedlabsi# systemctl start httpd
root@mohammedlabsi:/home/mohammedlabsi# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
root@mohammedlabsi:/home/mohammedlabsi#
```

Рисунок 4: Установка, запуск и добавление Apache HTTP Server в автозапуск

3.2 Просматриваю журнал ошибок Apache

```
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi# tail -f /var/log/httpd/error_log  
[Fri Oct 02 11:14:41.319862 2026] [suexec:notice] [pid 4496:tid 4496] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)  
AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::a00:27ff:fe80:d4e8%np0s3.  
Set the 'ServerName' directive globally to suppress this message  
[Fri Oct 02 11:14:41.329358 2026] [lbmethod_heartbeat:notice] [pid 4496:tid 4496] AH02282: No slotmem from mod_heartbeat  
[Fri Oct 02 11:14:41.329885 2026] [systemd:notice] [pid 4496:tid 4496] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0  
[Fri Oct 02 11:14:41.331014 2026] [mpm_event:notice] [pid 4496:tid 4496] AH00489: Apache/2.4.63 (Rocky Linux) configured -- resuming normal operations  
[Fri Oct 02 11:14:41.331021 2026] [core:notice] [pid 4496:tid 4496] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'
```

Рисунок 5: Просмотр стандартного журнала ошибок Apache

3.3 Настраиваю передачу сообщений через syslog

```
#
# EnableMMAP and EnableSendfile: On systems that support it,
# memory-mapping or the sendfile syscall may be used to deliver
# files. This usually improves server performance, but must
# be turned off when serving from networked-mounted
# filesystems or if support for these functions is otherwise
# broken on your system.
# Defaults if commented: EnableMMAP On, EnableSendfile Off
#
#EnableMMAP off
EnableSendfile on

# Supplemental configuration
#
# Load config files in the "/etc/httpd/conf.d" directory, if any.
IncludeOptional conf.d/*.conf

ErrorLog syslog:local1
```

1Help

2Save

3Mark

4Replac

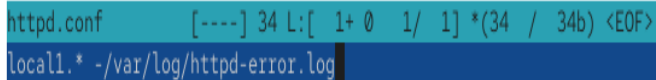
5Copy

6Move

7

Рисунок 6: Настройка Apache для передачи сообщений через syslog:local1

3.4 Создаю отдельный журнал Apache



```
httpd.conf      [----] 34 L:[ 1+ 0  1/ 1] *(34 / 34b) <EOF>  
local1.* -/var/log/httpd-error.log
```

Рисунок 7: Создание отдельного правила rsyslog для событий Apache

3.5 Перезапускаю службы

```
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi# systemctl restart rsyslog.service  
root@mohammedlabsi:/home/mohammedlabsi# systemctl restart httpd  
root@mohammedlabsi:/home/mohammedlabsi#
```

Рисунок 8: Перезапуск служб rsyslog и httpd после изменения конфигурации

3.6 Проверяю сообщения Apache

```
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi# tail -f /var/log/httpd/error_log  
[Fri Oct 02 11:14:41.319862 2026] [suexec:notice] [pid 4496:tid 4496] AH01232: suEXEC mechanism enabled (wrapper: /usr/sbin/suexec)  
AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::a00:27ff:fe80:d4e8%enp0s3.  
Set the 'ServerName' directive globally to suppress this message  
[Fri Oct 02 11:14:41.329358 2026] [lbmethod_heartbeat:notice] [pid 4496:tid 4496] AH02282: No slotmem from mod_heartbeat  
[Fri Oct 02 11:14:41.329885 2026] [systemd:notice] [pid 4496:tid 4496] SELinux policy enabled; httpd running as context system_u:system_r:httpd_t:s0  
[Fri Oct 02 11:14:41.331014 2026] [mpm_event:notice] [pid 4496:tid 4496] AH00489: Apache/2.4.63 (Rocky Linux) configured -- re  
suming normal operations  
[Fri Oct 02 11:14:41.331021 2026] [core:notice] [pid 4496:tid 4496] AH00094: Command line: '/usr/sbin/httpd -D FOREGROUND'  
[Fri Oct 02 11:19:26.945282 2026] [mpm_event:notice] [pid 4496:tid 4496] AH00492: caught SIGWINCH, shutting down gracefully
```

■

Рисунок 9: Сообщение Apache при перезапуске веб-службы

4. 4. Настройка отладочного журнала

4.1 Создаю файл debug.conf

```
root@mohammedlabsi:/etc/rsyslog.d#  
root@mohammedlabsi:/etc/rsyslog.d# cd /etc/rsyslog.d/  
root@mohammedlabsi:/etc/rsyslog.d# touch debug.conf  
root@mohammedlabsi:/etc/rsyslog.d# mcedit debug.conf  
  
root@mohammedlabsi:/etc/rsyslog.d# cat debug.conf  
*.debug /var/log/messages-debugroot@mohammedlabsi:/etc/rsyslog.d#  
root@mohammedlabsi:/etc/rsyslog.d#
```

Рисунок 10: Создание файла debug.conf для регистрации отладочных сообщений

4.2 Проверяю регистрацию debug-сообщений

```
.50.0 + 0x2a049)#012#0 0x0000000000000000 0/0 (/usr/vboxclient/.../2.10/00/vboxclient + 0x441a)#012ZLF 00j  
rchitecture: AMD x86-64  
Oct 2 11:22:24 mohammedlabsi systemd[1]: systemd-coredump@141-6591-0.service: Deactivated successfully.  
Oct 2 11:22:26 mohammedlabsi root[6597]: Daemon Debug Message  
Oct 2 11:22:26 mohammedlabsi root[6602]: Daemon Debug Message  
Oct 2 11:22:27 mohammedlabsi root[6604]: Daemon Debug Message  
Oct 2 11:22:29 mohammedlabsi kernel: traps: VBoxClient[6609] trap int3 ip:41e7eb sp:7f79b85b4cd0 error:0 in VBoxC  
400000+c8000]
```

Рисунок 11: Регистрация отладочного сообщения в отдельном журнале

5. 5. Работа с journalctl

5.1 Просматриваю системный журнал

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl
Oct 02 11:08:56 mohammedlabsi kernel: Linux version 6.12.0-211.53.1.el10_2.x86_64 (mockbuild@shou1-build-x86-01.releng.rockylinux.org) (gcc version 14.2.1 (Red Hat 14.2.1-1)) #1 SMP PREEMPT_DYNAMIC Mon Oct 1 11:08:56 CEST 2024
Oct 02 11:08:56 mohammedlabsi kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-211.53.1.el10_2.x86_64 root=/dev/mapper:rhel-root
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-provided physical RAM map:
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000009fbfff] usable
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x0000000000009fc000-0x0000000000009fffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x000000000000f00000-0x000000000000fffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x000000000001000000-0x000000000000dfffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000000dffff0000-0x00000000000dfffff] ACPI data
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000fec0000000-0x00000000fec00fff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000fee0000000-0x00000000fee00fff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000ffffc00000-0x00000000fffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x000000001000000000-0x0000000011fffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: NX (Execute Disable) protection: active
Oct 02 11:08:56 mohammedlabsi kernel: APIC: Static calls initialized
Oct 02 11:08:56 mohammedlabsi kernel: SMBIOS 2.5 present.
Oct 02 11:08:56 mohammedlabsi kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Oct 02 11:08:56 mohammedlabsi kernel: DMI: Memory slots populated: 0/0
Oct 02 11:08:56 mohammedlabsi kernel: Hypervisor detected: KVM
Oct 02 11:08:56 mohammedlabsi kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 02 11:08:56 mohammedlabsi kernel: kvm-clock: using sched offset of 5666333666 cycles
Oct 02 11:08:56 mohammedlabsi kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns:
Oct 02 11:08:56 mohammedlabsi kernel: tsc: Detected 3686.402 MHz processor
Oct 02 11:08:56 mohammedlabsi kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 02 11:08:56 mohammedlabsi kernel: e820: remove [mem 0x000a0000-0x000ffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: last_pfn = 0x120000 max_arch_pfn = 0x40000000
Oct 02 11:08:56 mohammedlabsi kernel: total RAM covered: 4096M
Oct 02 11:08:56 mohammedlabsi kernel: Found optimal setting for mtrr clean up
Oct 02 11:08:56 mohammedlabsi kernel: gran_size: 64K chunk_size: 1G num_reg: 3 lose cover RAM: 0G
Oct 02 11:08:56 mohammedlabsi kernel: MTRR map: 6 entries (3 fixed + 3 variable; max 35), built from 16 variable MTRRs
Oct 02 11:08:56 mohammedlabsi kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 02 11:08:56 mohammedlabsi kernel: e820: update [mem 0xe0000000-0xffffffff] usable ==> reserved
Oct 02 11:08:56 mohammedlabsi kernel: last_pfn = 0xe0000 max_arch_pfn = 0x40000000
Oct 02 11:08:56 mohammedlabsi kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
```

Рисунок 12: Просмотр системного журнала с помощью journalctl

5.2 Просматриваю журнал без пейджера

```
Oct 02 11:24:38 mohammedlabsl systemd[1]: Started systemd-coredump@167-6963-0.service - Process Core Dump (PID 6963/UID 0).
Oct 02 11:24:38 mohammedlabsl systemd-coredump[6964]: [?] Process 6959 (VBoxClient) of user 1000 dumped core.

Module /opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient without build-id.
Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64
Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64
Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64
Module libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64
Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64
Stack trace of thread 6962:
#0 0x00000000041e7eb n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x1e7eb)
#1 0x00000000041e764 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x1e764)
#2 0x000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x58b2c)
#3 0x00000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x3b900)
#4 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)
#5 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)

Stack trace of thread 6959:
#0 0x00007f79c6cfe25d syscall (libc.so.6 + 0x10425d)
t + 0x3a812)
#1 0x00000000043a812 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x58786)
#2 0x000000000458786 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x56d8)
#3 0x0000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
#4 0x00007f79c6c2458e __libc_start_call_main (libc.so.6 + 0x2a58e)
#5 0x00007f79c6c24649 __libc_start_main@@@GLIBC_2.34 (libc.so.6 + 0x2a649)
9)
#6 0x0000000004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
t + 0x44fa)

ELF object binary architecture: AMD x86-64
Oct 02 11:24:38 mohammedlabsl systemd[1]: systemd-coredump@167-6963-0.service: Deactivated successfully.
root@mohammedlabsl:/home/mohammedlabsl#
```

Рисунок 13: Просмотр journalctl без использования пейджера

5.3 Запускаю journalctl в реальном времени

```
t + 0x3b900)                                #3 0x00000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

#4 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)
#5 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)

Stack trace of thread 7001:
#0 0x00007f79c6cfe25d syscall (libc.so.6 + 0x10425d)
#1 0x00000000043af60 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x3af60)                                #2 0x00000000045930b n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x5930b)                                #3 0x00000000043b99a n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x3b99a)                                #4 0x000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x58b2c)                                #5 0x00000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x3b900)                                #6 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)
#7 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)

Stack trace of thread 7000:
#0 0x00007f79c6cfe25d syscall (libc.so.6 + 0x10425d)
#1 0x00000000043a812 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x3a812)                                #2 0x000000000458786 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x58786)                                #3 0x0000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x56d8)                                #4 0x00007f79c6c2458e __libc_start_call_main (libc.so.6 + 0x2a58e)
#5 0x00007f79c6c24649 __libc_start_main@@GLIBC_2.34 (libc.so.6 + 0x2a64

9)                                           #6 0x0000000004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxCli

t + 0x44fa)                                ELF object binary architecture: AMD x86-64

Oct 02 11:24:53 mohammedlabsi systemd[1]: systemd-coredump@170-7004-0.service: Deactivated successfully.
```

Рисунок 14: Мониторинг системного журнала journalctl в реальном времени

5.4 Изучаю параметры фильтрации

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl
Display all 129 possibilities? (y or n)

_AUDIT_LOGINUID=      CPU_USAGE_NSEC=      _PID=
_AUDIT_SESSION=      CURRENT_USE=         PODMAN_EVENT=
AVAILABLE=           CURRENT_USE_PRETTY=  PODMAN_TIME=
AVAILABLE_PRETTY=    DBUS_BROKER_LOG_DROPPED=  PODMAN_TYPE=
_BOOT_ID=            DBUS_BROKER_METRICS_DISPATCH_AVG=  PRIORITY=
_CAP_EFFECTIVE=      DBUS_BROKER_METRICS_DISPATCH_COUNT=  REALMD_OPERATION=
_CMDLINE=            DBUS_BROKER_METRICS_DISPATCH_MAX=  _RUNTIME_SCOPE=
CODE_FILE=           DBUS_BROKER_METRICS_DISPATCH_MIN=  SEAT_ID=
CODE_FUNC=           DBUS_BROKER_METRICS_DISPATCH_STDDEV=  _SELINUX_CONTEXT=
CODE_LINE=           DISK_AVAILABLE=      SESSION_ID=
_COMM=              DISK_AVAILABLE_PRETTY=  _SOURCE_BOOTTIME_TIMESTAMP=
CONFIG_FILE=         DISK_KEEP_FREE=      _SOURCE_MONOTONIC_TIMESTAMP=
CONFIG_LINE=         DISK_KEEP_FREE_PRETTY=  _SOURCE_REALTIME_TIMESTAMP=
COREDUMP_BY_PIDFD=   ERRNO=              SSSD_DOMAIN=
COREDUMP_CGROUP=     _EXE=              SSSD_PRG_NAME=
COREDUMP_CMDLINE=    _GID=              _STREAM_ID=
COREDUMP_COMM=       GLIB_DOMAIN=        SYSLOG_FACILITY=
COREDUMP_CWD=        GLIB_OLD_LOG_API=   SYSLOG_IDENTIFIER=
COREDUMP_DUMPABLE=   _HOSTNAME=          SYSLOG_PID=
COREDUMP_ENVIRON=    INITRD_USEC=        SYSLOG_RAW=
COREDUMP_EXE=        INVOCATION_ID=      SYSLOG_TIMESTAMP=
COREDUMP_FILENAME=   JOB_ID=             _SYSTEMD_CGROUP=
COREDUMP_GID=        JOB_RESULT=         _SYSTEMD_INVOCATION_ID=
COREDUMP_HOSTNAME=   JOB_TYPE=           _SYSTEMD_OWNER_UID=
COREDUMP_OPEN_FDS=   JOURNAL_NAME=       _SYSTEMD_SESSION=
COREDUMP_OWNER_UID=  JOURNAL_PATH=       _SYSTEMD_SLICE=
COREDUMP_PACKAGE_JSON=  _KERNEL_DEVICE=    _SYSTEMD_UNIT=
```

Рисунок 15: Просмотр доступных параметров фильтрации journalctl

5.5 Фильтрую события по UID

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl _UID=0
Oct 02 11:08:56 mohammedlabsi systemd-journald[291]: Collecting audit messages is disabled.
Oct 02 11:08:56 mohammedlabsi systemd-journald[291]: Journal started
Oct 02 11:08:56 mohammedlabsi systemd-journald[291]: Runtime Journal (/run/log/journal/4c126675941a4100b4e72d46c37d47d9) is 8
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Inserted module 'i2c_dev'
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Module 'msr' is built in
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Inserted module 'fuse'
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Module 'scsi_dh_alua' is built in
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Module 'scsi_dh_emc' is built in
Oct 02 11:08:56 mohammedlabsi systemd-modules-load[293]: Module 'scsi_dh_rdac' is built in
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating group 'nobody' with GID 65534.
Oct 02 11:08:56 mohammedlabsi systemd[1]: Finished systemd-sysctl.service - Apply Kernel Variables.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating group 'users' with GID 100.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating group 'systemd-journal' with GID 190.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating group 'dbus' with GID 81.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating user 'dbus' (System Message Bus) with UID 81 and GID 81.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating group 'tss' with GID 59.
Oct 02 11:08:56 mohammedlabsi systemd-sysusers[305]: Creating user 'tss' (Account used for TPM access) with UID 59 and GID 59.
Oct 02 11:08:56 mohammedlabsi systemd[1]: Finished systemd-sysusers.service - Create System Users.
Oct 02 11:08:56 mohammedlabsi systemd[1]: Starting systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev...
Oct 02 11:08:56 mohammedlabsi systemd[1]: Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Oct 02 11:08:56 mohammedlabsi systemd-tmpfiles[319]: /usr/lib/tmpfiles.d/var.conf:14: Duplicate line for path "/var/log", ign
Oct 02 11:08:56 mohammedlabsi systemd[1]: Finished systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev.
Oct 02 11:08:56 mohammedlabsi systemd[1]: Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Oct 02 11:08:56 mohammedlabsi systemd[1]: Finished systemd-vconsole-setup.service - Virtual Console Setup.
Oct 02 11:08:56 mohammedlabsi systemd[1]: dracut-cmdline-ask.service - dracut ask for additional cmdline parameters was skippe
Oct 02 11:08:56 mohammedlabsi systemd[1]: Starting dracut-cmdline.service - dracut cmdline hook...
Oct 02 11:08:57 mohammedlabsi dracut-cmdline[325]: dracut-107-9.el10_2
Oct 02 11:08:57 mohammedlabsi dracut-cmdline[325]: Using kernel command line parameters: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.1
Oct 02 11:08:57 mohammedlabsi systemd[1]: Finished dracut-cmdline.service - dracut cmdline hook.
```

Рисунок 16: Фильтрация журнала journalctl по UID суперпользователя

5.6 Просматриваю последние записи журнала

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl -n 20
Oct 02 11:25:44 mohammedlabsi kernel: traps: VBoxClient[7155] trap int3 ip:41e7eb sp:7f79b85b4cd0 error:0 in VBoxClient[1e7eb
Oct 02 11:25:44 mohammedlabsi systemd-coredump[7156]: Process 7152 (VBoxClient) of user 1000 terminated abnormally with signal
Oct 02 11:25:44 mohammedlabsi systemd[1]: Started systemd-coredump@180-7156-0.service - Process Core Dump (PID 7156/UID 0).
Oct 02 11:25:44 mohammedlabsi systemd-coredump[7157]: [..] Process 7152 (VBoxClient) of user 1000 dumped core.

                                Module /opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient without build-id.
                                Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64
                                Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64
                                Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64
                                Module libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64
                                Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64
                                Stack trace of thread 7155:
                                #0 0x00000000041e7eb n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #1 0x00000000041e764 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #2 0x000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #3 0x00000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #4 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)
                                #5 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)

                                Stack trace of thread 7152:
                                #0 0x00007f79c6cfe25d syscall (libc.so.6 + 0x10425d)
                                #1 0x00000000043a812 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #2 0x000000000458786 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #3 0x0000000004056d8 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                #4 0x00007f79c6c2458e __libc_start_call_main (libc.so.6 + 0x2a58e)
                                #5 0x00007f79c6c24649 __libc_start_main@@GLIBC_2.34 (libc.so.6 + 0x2a649)
                                #6 0x0000000004044fa n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                                ELF object binary architecture: AMD x86-64

Oct 02 11:25:44 mohammedlabsi systemd[1]: systemd-coredump@180-7156-0.service: Deactivated successfully.
```

Рисунок 17: Просмотр последних двадцати записей системного журнала

5.7 Фильтрую сообщения по уровню ошибки

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl -p err
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: cnic
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: cnic_init
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: bnx2i
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: bnx2i_mod_init
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: e1000
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: e1000_init_module
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervi
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid
Oct 02 11:09:02 mohammedlabsi alsactl[1161]: alsa-lib main.c:1804:(snd_use_case_mgr_open) [error.ucm] failed to import hw:0 u
Oct 02 11:10:08 mohammedlabsi gdm-password[2194]: gkr-pam: unable to locate daemon control file
Oct 02 11:10:13 mohammedlabsi gdm-password[2206]: gkr-pam: unable to locate daemon control file
Oct 02 11:10:22 mohammedlabsi systemd[2215]: Failed to start app-gnome-user\x2ddirs\x2dupdate\x2dgtk-2494.scope - Application
Oct 02 11:10:23 mohammedlabsi systemd-coredump[3022]: [ ] Process 2993 (VBoxClient) of user 1000 dumped core.

Module /opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient without build-id.
Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64
Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64
Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64
Module libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64
Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64
Stack trace of thread 3000:
#0 0x0000000000041e7b n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
#1 0x0000000000041e74 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
#2 0x00000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
#3 0x0000000000043b900 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
#4 0x00007f79c6c8f8bf start_thread (libc.so.6 + 0x958bf)
#5 0x00007f79c6d0043c __clone3 (libc.so.6 + 0x10643c)
```

Рисунок 18: Фильтрация journalctl по сообщениям уровня error

5.8 Просматриваю события за период

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl --since yesterday
Oct 02 11:08:56 mohammedlabsi kernel: Linux version 6.12.0-211.53.1.el10_2.x86_64 (mockbuild@houl-build-x86-01.releng.rockylin
Oct 02 11:08:56 mohammedlabsi kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-211.53.1.el10_2.x86_64 root=/dev/mapp
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-provided physical RAM map:
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000000dfffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x000000000dff0000-0x000000000dffffff] ACPI data
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
Oct 02 11:08:56 mohammedlabsi kernel: BIOS-e820: [mem 0x0000000100000000-0x000000011fffffffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: NX (Execute Disable) protection: active
Oct 02 11:08:56 mohammedlabsi kernel: APIC: Static calls initialized
Oct 02 11:08:56 mohammedlabsi kernel: SMBIOS 2.5 present.
Oct 02 11:08:56 mohammedlabsi kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
Oct 02 11:08:56 mohammedlabsi kernel: DMI: Memory slots populated: 0/0
Oct 02 11:08:56 mohammedlabsi kernel: Hypervisor detected: KVM
Oct 02 11:08:56 mohammedlabsi kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
Oct 02 11:08:56 mohammedlabsi kernel: kvm-clock: using sched offset of 5666333666 cycles
Oct 02 11:08:56 mohammedlabsi kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns
Oct 02 11:08:56 mohammedlabsi kernel: tsc: Detected 3686.402 MHz processor
Oct 02 11:08:56 mohammedlabsi kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
Oct 02 11:08:56 mohammedlabsi kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
Oct 02 11:08:56 mohammedlabsi kernel: last_pfn = 0x120000 max_arch_pfn = 0x40000000
Oct 02 11:08:56 mohammedlabsi kernel: total RAM covered: 4096M
Oct 02 11:08:56 mohammedlabsi kernel: Found optimal setting for mtrr clean up
Oct 02 11:08:56 mohammedlabsi kernel: gran_size: 64K chunk_size: 1G num_reg: 3 lose cover RAM: 0G
Oct 02 11:08:56 mohammedlabsi kernel: MTRR map: 6 entries (3 fixed + 3 variable; max 35), built from 16 variable MTRRs
Oct 02 11:08:56 mohammedlabsi kernel: x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
Oct 02 11:08:56 mohammedlabsi kernel: e820: update [mem 0xe0000000-0xfffffffff] usable ==> reserved
Oct 02 11:08:56 mohammedlabsi kernel: last_pfn = 0xe0000 max_arch_pfn = 0x400000000
Oct 02 11:08:56 mohammedlabsi kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
```

Рисунок 19: Просмотр событий journalctl начиная со вчерашнего дня

5.9 Просматриваю ошибки за период

```
root@mohammedlabsi:/home/mohammedlabsi# journalctl --since yesterday -p err
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: cnic
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: cnic_init
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: bnx2i
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: bnx2i_mod_init
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: e1000
Oct 02 11:08:57 mohammedlabsi kernel: Warning: Unmaintained driver is detected: e1000_init_module
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervi
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
Oct 02 11:08:57 mohammedlabsi kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid
Oct 02 11:09:02 mohammedlabsi alsactl[1161]: alsa-lib main.c:1804:(snd_use_case_mgr_open) [error.ucm] failed to import hw:0 u
Oct 02 11:10:08 mohammedlabsi gdm-password[2194]: gkr-pam: unable to locate daemon control file
Oct 02 11:10:13 mohammedlabsi gdm-password[2206]: gkr-pam: unable to locate daemon control file
Oct 02 11:10:22 mohammedlabsi systemd[2215]: Failed to start app-gnome-user\x2ddirs\x2dupdate\x2dgtk-2494.scope - Application
Oct 02 11:10:23 mohammedlabsi systemd-coredump[3022]: [?] Process 2993 (VBoxClient) of user 1000 dumped core.

                               Module /opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient without build-id.
                               Module libXau.so.6 from rpm libXau-1.0.11-8.el10.x86_64
                               Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64
                               Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64
                               Module libffi.so.8 from rpm libffi-3.4.4-10.el10.x86_64
                               Module libwayland-client.so.0 from rpm wayland-1.24.0-1.el10.x86_64
                               Stack trace of thread 3000:
                               #0  0x00000000041e7eb n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                               #1  0x00000000041e764 n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
                               #2  0x000000000458b2c n/a (/opt/VBoxGuestAdditions-7.2.16/bin/VBoxClient)
```

Рисунок 20: Просмотр ошибок journalctl, зарегистрированных начиная со вчерашнего дня

5.10 Используя подробный формат вывода

```
root@mohammedlabsi: /home/mohammedlabsi
root@mohammedlabsi: /home/mohammedlabsi# journalctl -o verbose
Fri 2026-10-02 11:08:56.885102 MSK [s=9060f493cde14945bec7d9c771455942;i=1;b=f5ec7366607645bd84f5ad592976373a;m=d4b39;t=65cd78]
  _SOURCE_BOOTTIME_TIMESTAMP=0
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  PRIORITY=5
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  MESSAGE=Linux version 6.12.0-211.53.1.el10_2.x86_64 (mockbuild@h01-build-x86-01.releng.rockylinux.org) (gcc (GCC) 14.3.1)
  _BOOT_ID=f5ec7366607645bd84f5ad592976373a
  _MACHINE_ID=4c126675941a4100b4e72d46c37d47d9
  _HOSTNAME=mohammedlabsi
  _RUNTIME_SCOPE=initrd
Fri 2026-10-02 11:08:56.885131 MSK [s=9060f493cde14945bec7d9c771455942;i=2;b=f5ec7366607645bd84f5ad592976373a;m=d4b56;t=65cd78]
  _SOURCE_BOOTTIME_TIMESTAMP=0
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  _BOOT_ID=f5ec7366607645bd84f5ad592976373a
  _MACHINE_ID=4c126675941a4100b4e72d46c37d47d9
  _HOSTNAME=mohammedlabsi
  _RUNTIME_SCOPE=initrd
  PRIORITY=6
  MESSAGE=Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-211.53.1.el10_2.x86_64 root=/dev/mapper/rl-root ro resume=UUID=
Fri 2026-10-02 11:08:56.885138 MSK [s=9060f493cde14945bec7d9c771455942;i=3;b=f5ec7366607645bd84f5ad592976373a;m=d4b5d;t=65cd78]
  _SOURCE_BOOTTIME_TIMESTAMP=0
  _SOURCE_MONOTONIC_TIMESTAMP=0
  _TRANSPORT=kernel
  SYSLOG_FACILITY=0
  SYSLOG_IDENTIFIER=kernel
  _BOOT_ID=f5ec7366607645bd84f5ad592976373a
  _MACHINE_ID=4c126675941a4100b4e72d46c37d47d9
  _HOSTNAME=mohammedlabsi
```

Рисунок 21: Детальный формат вывода журнала journalctl

5.11 Просматриваю журнал службы sshd

```
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi#  
root@mohammedlabsi:/home/mohammedlabsi# journalctl _SYSTEMD_UNIT=sshd.service  
Oct 02 11:09:03 mohammedlabsi sshd[1285]: Server listening on 0.0.0.0 port 22.  
Oct 02 11:09:03 mohammedlabsi sshd[1285]: Server listening on :: port 22.  
root@mohammedlabsi:/home/mohammedlabsi#
```

Рисунок 22: Просмотр сообщений службы sshd с помощью journalctl

6. 6. Постоянный журнал journald

6.1 Настраиваю постоянное хранение журнала

```
root@mohammedlabsl:/home/mohammedlabsl#  
root@mohammedlabsl:/home/mohammedlabsl# mkdir -p /var/log/journal  
root@mohammedlabsl:/home/mohammedlabsl# chown root:systemd-journal /var/log/journal/  
root@mohammedlabsl:/home/mohammedlabsl# chmod 2755 /var/log/journal/  
root@mohammedlabsl:/home/mohammedlabsl# killall -USR1 systemd-journald  
root@mohammedlabsl:/home/mohammedlabsl#  
root@mohammedlabsl:/home/mohammedlabsl# journalctl -b  
Oct 02 11:08:56 mohammedlabsl kernel: Linux version 6.12.0-211.53.1.el10_2.x86_64 (mockbuild@h0u1-build-x86-01.releng.rockylin  
Oct 02 11:08:56 mohammedlabsl kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-6.12.0-211.53.1.el10_2.x86_64 root=/dev/mapp  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-provided physical RAM map:  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x0000000000100000-0x0000000000dfffff] usable  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x0000000000dff0000-0x0000000000dfffff] ACPI data  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0ffff] reserved  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee0ffff] reserved  
Oct 02 11:08:56 mohammedlabsl kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
```

Рисунок 23: Настройка постоянного хранения journald и просмотр журнала текущей загрузки

7. 7. Выводы по проделанной работе

В ходе лабораторной работы были изучены средства управления системными журналами Linux.

Был освоен мониторинг событий в реальном времени с помощью **tail -f**, а также регистрация пользовательских сообщений командой **logger**.

Была выполнена настройка **rsyslog** для перенаправления сообщений веб-службы Apache в отдельный файл журнала и создан отдельный журнал для отладочных сообщений.

Также были изучены основные возможности команды **journalctl**: просмотр журнала, работа в режиме реального времени, фильтрация по UID, приоритету, времени и системной службе.

В завершение было настроено постоянное хранение журнала **systemd-journald** в каталоге **/var/log/journal**.